GRADE 10 · MATHEMATICS ENGLISH MEDIUM

QUESTION PAPER

Model Paper — 01

Third Term Test (Full Syllabus)

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Time: 3 hours **Total marks:** 100

Name: ..... Class: ..... Date: .....

Instructions:

- This question paper consists of two parts, **A** and **B**. Answer **all** the questions.
- Answer the 20 questions in **Part A** in the space provided on this paper itself. (2 marks for each question)
- Answer the 6 questions in **Part B** on the separate sheets of paper provided. (10 marks for each question)
- Show the steps of your calculations clearly. Draw graphs on the graph paper provided.
- Where necessary, take the value of π as $\frac{22}{7}$. The use of calculators is not allowed.

Part	A	B1	B2	B3	B4	B5	B6	Total
Marks	/40	/10	/10	/10	/10	/10	/10	/100

Part A

Answer all the questions in the space provided on this paper. (2 × 20 = 40 marks)

1. Find the square root of 2304 using the division method.

Answer: .....

2. Simplify: $2\frac{1}{3} \div 1\frac{1}{6} \times \frac{3}{4}$

Answer: .....

3. Expand: $(2x - 3)^2$

Answer: .....

4. Factorise: $6x^2 - x - 2$

Answer: .....

5. Find the least common multiple of the algebraic expressions $4a^2b$, $6ab^3$ and $9a^3$.

Answer: .....

6. Simplify: $\frac{3}{x - 2} - \frac{2}{x + 1}$

Answer: .....

7. Find the perimeter of a sector of a circle of radius 21 cm and centre angle 120°.

Answer: .....

8. In triangle ABC, AB = AC and $B\hat{A}C = 40°$. If the side BC is produced to D, find the magnitude of $A\hat{C}D$.

Answer: .....

9. Find the value of: $\log_2 32 + \log_3 \frac{1}{9}$

Answer: .....

10. If $\log_x 81 = 4$, find the value of x.

Answer: .....

11. Find the 15th term of an arithmetic progression whose first term is 7 and common difference is 4.

Answer: .....

12. For two sets A and B, n(A) = 18, n(B) = 15 and $n(A \cup B) = 27$. Find $n(A \cap B)$.

Answer: .....

13. Find the interest received when a sum of Rs. 25 000 is deposited for $2\frac{1}{2}$ years at an annual simple interest rate of 12%.

Answer: .....

14. A piece of work can be completed by 12 workers in 15 days. Find the number of days taken to complete it by 18 workers working at the same rate.

Answer: .....

15. Make a the subject of the formula $v^2 = u^2 + 2as$.

Answer: .....

16. Solve the inequality $2x - 5 < 3x + 1$ and write the least integer value that x can take.

Answer: .....

17. Of 72 students in a class, 20 like playing cricket. When this information is shown in a pie chart, find the centre angle for cricket.

Answer: .....

18. A bag contains 4 red balls and 6 blue balls of the same size. When one ball is taken at random, find the probability that it is a red ball.

Answer: .....

19. In the parallelogram ABCD, $\hat{A} = (2x + 10)°$ and $\hat{B} = (3x - 5)°$. Find the magnitude of $\hat{C}$.

Answer: .....

20. Find the volume of a solid cylinder of radius 7 cm and height 10 cm.

Answer: .....

Part B

Answer all the questions on separate sheets of paper. (10 × 6 = 60 marks)

1. (a) Solve: $3x + 2y = 16$, $5x - 3y = 14$ (4 marks)

(b) Solve using factors: $x^2 - 5x - 14 = 0$ (2 marks)

(c) The length of a rectangular plot of land is 3 m more than its width. Its area is 54 m^2.

(i) Taking the width of the plot as x m, form a quadratic equation in terms of x. (1 mark)

(ii) Solve that equation and find the length and the width of the plot. (2 marks)

(iii) Find the length of wire needed to go around the plot 3 times. (1 mark)

2. An incomplete table of values prepared for drawing the graph of the function $y = x^2 - 4$ is given below.

x	−3	−2	−1	0	1	2	3
y	5	.....	−3	−4	.....	0	5

(a) Complete the blanks in the table. (1 mark)

(b) Draw the graph taking 10 small squares along the x-axis to represent 1 unit and 10 small squares along the y-axis to represent 1 unit. (3 marks)

(c) Using the graph,

(i) write the minimum value of the function. (1 mark)

(ii) write the equation of the axis of symmetry of the graph. (1 mark)

(iii) write the range of values of x for which y is not positive. (1 mark)

(iv) find the roots of the equation $x^2 - 6 = 0$ and, using them, write an approximate value for $\sqrt{6}$. (2 marks)

(d) Write the function of the graph obtained when the graph is moved up by 3 units along the y-axis. (1 mark)

3. A cuboid-shaped tank with internal dimensions of length 2.2 m, width 1.4 m and height 0.5 m is completely full of water. (1 m^3 = 1000 litres)

(a) Find the volume of water in the tank in m^3 and write how many litres it is. (2 marks)

(b) When all of this water is poured into a cylindrical tank of internal radius 70 cm, find the height of the water level in it. (3 marks)

(c) Find the area of the curved surface of the cylindrical tank that is in contact with the water. (2 marks)

(d) If the height of the cylindrical tank is 1.5 m, find the number of litres of water still needed to fill it completely. Find the time taken to supply that water through a pipe that supplies 35 litres of water per minute. (3 marks)

4. The diagonals AC and BD of the parallelogram ABCD intersect at O. A straight line drawn through O meets the side AB at P and the side DC at Q.

(a) Draw a rough sketch including the above information. (1 mark)

(b) Prove that the triangles AOP and COQ are congruent. (4 marks)

(c) Show that AP = CQ. (1 mark)

(d) Prove that the quadrilateral PBQD is a parallelogram. (2 marks)

(e) If the area of the parallelogram ABCD is 60 cm^2, find the area of the triangle AOB. (2 marks)

5. The following frequency distribution shows the marks obtained by 40 students in a mathematics test. (10 – 20 means 10 marks or more and less than 20.)

Marks	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60
Number of students	4	7	12	10	7

(a) Write the modal class. (1 mark)

(b) Using the class mid-values, find the mean mark of the students. (4 marks)

(c) Find the percentage of students who obtained 40 marks or more. (2 marks)

(d) Later, another 10 students sat the same test and their mean mark was 47.25. Find the mean mark of all 50 students. (3 marks)

6. In a class of 40 students, 22 play cricket and 18 play volleyball. There are 7 students who play neither of the two sports.

(a) Find the number of students who play both sports. (2 marks)

(b) Show this information in a Venn diagram. (3 marks)

(c) Write the number of students who play cricket only. (1 mark)

(d) Find the probability that a student selected at random from the class

(i) plays only one of the two sports, (2 marks)

(ii) does not play volleyball. (2 marks)

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